MIL-G-1086E
21 February 1975
SUPERSEDING
MIL-G-1086D
15 October 1968

MILITARY SPECIFICATION

GASKET MATERIAL, SYNTHETIC RUBBER

(FOR BOLTED STEEL TANKS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of gasket material manufactured of synthetic rubber.

1.2 Classification. The gasket material shall be of the following types, as specified (see 6.2):

Type I - Sheet (includes strip, flange, bolt, or washer ring).
Type II - Molded (includes wedge-shaped and fillet bending-radii gasket).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-576	- Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

FSC 5330

Source: https://assist.dla.mil -- Downloaded: 202640-16T12:40Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-190	- Identification Marking of Rubber Products.
MIL-STD-289	- Visual Inspection Guide for Rubber Sheet Material.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395	- Compression Set of Vulcanized Rubber.
D 412	- Tension Testing of Vulcanized Rubber.
D 471	- Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids.
D 573	- Accelerated Aging of Vulcanized Rubber by the Oven Method.
D 1229	- Low-Temperature Compression Set of Vulcanized Elastomers.
D 2240	- Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:40Z
Check the source to verify that this is the current version before use.

Naval Shipboard

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The gasket material shall consist of synthetic rubber in sheet or molded form as specified herein.

3.2 Physical properties. The physical properties of the gasket material shall conform to Table I when tested in accordance with Table III.

Table I. Physical Requirements

Description	Value Type I	Value Type II
Tensile strength, initial, psi minimum.	1200	1500
Tensile strength after oven aging, percent of initial, minimum.	70	70
Tensile strength after immersion in ASTM D 471 Fuel B, percent of initial, minimum.	50	50
Tensile strength after immersion in distilled water, percent of initial, minimum.	60	60
Ultimate elongation, initial, percent, minimum.	175	500
Ultimate elongation after oven aging, percent of initial, minimum.	70	70
Hardness, Shore A.	75 ± 5	50 ± 5
Hardness change, Shore A, after oven aging, maximum.	7	7
Compression set, maximum percent of original, after oven aging.	40	50
Low-temperature compression set, maximum (percent of original).	60	60
Weight change after immersion in ASTM D 471 Fuel A and air-dry (maximum) (percent).	10	10
Volume change after immersion in ASTM D 471 Fuel B (maximum) (percent).	70	70

3.3 Size. The size of the finished gasket material shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 20260-16T12:40Z
Check the source to verify that this is the current version before use.

3.4 Tolerance. Tolerance in dimension shall be as follows:

Thickness:
Plus or minus 1/64 inch, unless otherwise specified (see 6.2).

Width:
Plus or minus 1/16 inch.

Length:
Type I, sheet (sheet gasket), plus or minus 3 inches.
Type I, sheet (strip), plus 3 feet or minus 1 foot.
Type II, molded gasket, plus or minus 1/16 inch.

3.5 Identification marking. The gasket material shall be marked for identification in accordance with MIL-STD-190.

3.6 Workmanship. Rubber gasket material shall be free of pockmarks, material overlap, surface voids, moisture pockets, foreign material, or surface lumps.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.4).

4.3 Quality conformance inspection.

4.3.1 Inspection lot. For the purpose of inspection, all gasket material of the same type offered for delivery at one time shall be considered one lot.

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:40Z
Check the source to verify that this is the current version before use.

4.3.2 Examination. Sampling for examination shall be in accordance with MIL-STD-105. Samples shall be examined as specified in MIL-STD-289. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Tests. Samples shall be selected from each lot offered as specified in Table II. Samples of Type I gasket material shall be taken from the sheet after the sheet is cured and before the gaskets are cut and punched to size. Samples of Type II gasket material shall be warranted to have equivalent cure and to be from the same run or batch of compound used in the lot offered. The gasket material shall be tested in accordance with Table III. Nonconformance to 3.2 shall be cause for rejection.

Table II. Test Samples

Gasket Material Type	Gasket Material Description	Sample Quantity and Description
I	Strip and flange	Two samples, 24 in. x 24 in., cut from sheet.
I	Washer ring, bolt and sheet	Two samples, 24 in. x 24 in., cut from sheet.
II	Wedge-shaped	Three samples, 6 in. x 6 in. x 0.080 ± 0.010 in., and four samples, 2 in. x 8 in. x 0.50 ± 0.10 in., molded from representative run or batch material.
II	Fillet bending-radii	Four samples, 2 in. x 8 in. x 0.50 ± 0.10 in., and three samples, 6 in. x 6 in. x 0.080 ± 0.010 in., molded from representative run or batch material.

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:40Z
Check the source to verify that this is the current version before use.

Table III. Material Tests

Description	ASTM Standards Designation
Tensile strength, initial.	D 412
Tensile strength after oven aging.	1/D 573
Tensile strength after immersion in reference Fuel B.	2/D 471
Tensile strength after immersion in distilled water.	3/D 471
Ultimate elongation, initial.	D 412
Ultimate elongation after oven aging.	1/D 573
Hardness, Shore A, initial.	D 2240
Hardness, Shore A, after oven aging.	1/D 573
Compression set after oven aging.	1/D 395
Compression set at minus 20° F.	4/D 1229
Weight change in reference Fuel A.	5/D 471
Volume change in reference Fuel B.	6/D 471

1/ Test specimens shall be conditioned at 194° F ± 2° F for 46 hours ± 1/2 hour.
2/ Samples shall be immersed for 48 hours ± 1 hour at 75° F ± 5° F.
3/ Specimen shall be immersed not less than 1 hour at 212° F ± 2° F.
4/ Specimen shall be aged at minus 20° F ± 2° F for 94 hours. Compression set shall be calculated 30 minutes after release from compression. Method B shall be used.
5/ Specimen shall be immersed in reference Fuel A for 46 hours ± 1/4 hour at 75° F ± 5° F. After the immersion period, the specimen shall be removed from the fuel and airdried for a minimum of 24 hours at 75° F ± 5° F before determining weight after test W_2.
6/ Specimen shall be immersed in reference Fuel B for 46 hours ± 1/4 hour at 75° F ± 5° F.

4.4 Inspection preparation for delivery.

4.4.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:40Z
Check the source to verify that this is the current version before use.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Materials, methods and containers not as specified. Each incorrect material, method or container shall be considered one defect.
102.	Gross weight of boxes exceeds the limitations given in the box specification for Level A.
103.	Individual packages do not contain gasket material of like size and description for Level A.
104.	Strip and sheet gasket material not rolled and secured as specified for Level A.
105.	Flange gasket material folded for Level A shipments.
106.	Fiberboard boxes not waterproof sealed for Level A or B.
107.	Marking illegible, incorrect or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Ring, wedge, and radii gasket material. Gasket material of like size and description shall be packaged in close-fitting fiberboard boxes conforming to PPP-B-636, V3c or V3s, style optional, in quantities not to exceed the gross weight limitations of the box specification. The boxes shall be waterproof sealed in accordance with the appendix to the box specification.

5.1.1.2 Strip and sheet gasket material. Strip gasket material of like size and description shall be individually rolled to a minimum safe diameter and the rolls secured with cotton twine. Sheet gasket material shall be rolled together to a minimum safe diameter and secured in at least two places with cotton twine. After being rolled as specified, the gasket material shall be packaged in close-fitting fiberboard boxes conforming to PPP-B-636, V3c or V3s, style optional, in quantities not to exceed the gross weight limitations of the box specification. The boxes shall be waterproof sealed in accordance with the appendix to the box specification.

5.1.1.3 Flange gasket material. Flange gasket material of like size and description, shall be packaged without folding in close-fitting fiberboard boxes conforming to PPP-B-636, V3c or V3s, style optional, in quantities not to exceed the gross weight limitations of the box specification. The boxes shall be waterproof sealed in accordance with the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:40Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The gasket material shall be packaged in a manner to afford protection against damage during shipment from the supplier to the initial destination. The supplier's standard practice will be acceptable provided it fulfills these requirements.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. Gasket material of like size and description, packaged as specified, shall be packed in boxes conforming to PPP-B-601, Overseas Type, Grade B, or PPP-B-621, Class 2.

5.2.2 Level B. Gasket material of like size and description, packaged as specified for Level A, shall be packed in Domestic Type boxes conforming to PPP-B-576, PPP-B-601, Grade B, or PPP-B-621, or in weather-resistant boxes conforming to PPP-B-640. Fiberboard boxes shall be waterproof sealed in accordance with the box specification appendix.

5.2.3 Level C. Gasket material shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I sheet gasket material is for flat sealing surfaces in bolted steel tanks.

6.1.2 Type II. Type II molded gasket material is for grooved casting facing and other specially shaped sealing areas.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of gasket material required (see 1.2).
(c) Size and shape of gasket material required (see 3.3).
(d) When thickness tolerance is other than as specified (see 3.4).
(e) Level of packaging and level of packing required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-G-1086E Gasket Material, Synthetic Rubber (For Bolted Steel Tanks)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 20261O-16T12:40Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Center
ATTN: STSFB-DS
Fort Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:40Z
Check the source to verify that this is the current version before use.